

A S S E S S M E N T 3 — I M P L E M E N T A T I O N

NYC Taxi Fare Prediction

From Proposal to Production: Implementing GBDT for Dynamic Pricing & Fleet Optimization

Gradient Minds — Team 4

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Meet the Team!

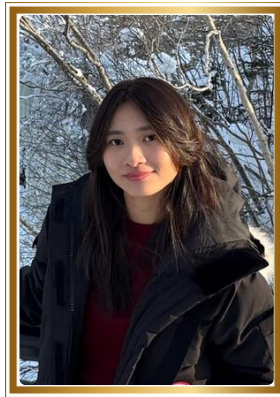


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Business Challenge

Why NYC TLC needs a data-driven transformation?

1 Fare Prediction Inefficiency

Current State:

Flat fixed pricing — no surge

Business Cost:

Lost revenue on peak trips

2 Inefficient Taxi Allocation

Current State:

Drivers guess hotspots

Business Cost:

30%+ idle time, fuel waste

3 Demand-Supply Mismatch

Current State:

No peak/off-peak strategy

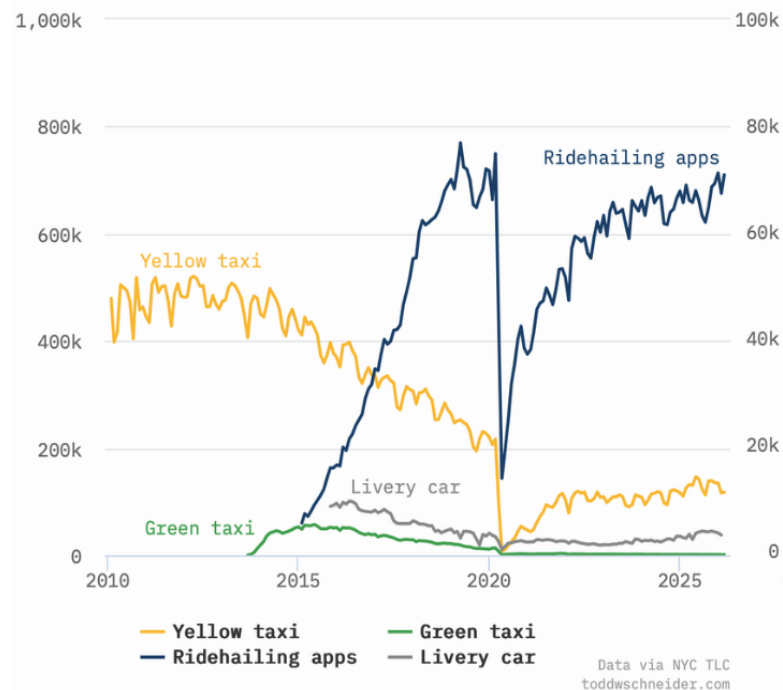
Business Cost:

Customer churn to Uber/Lyft

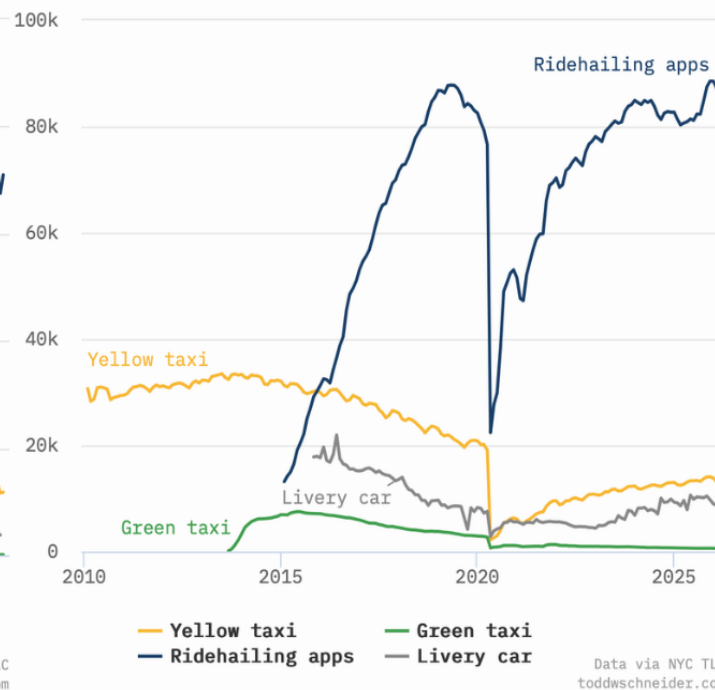
Strategic Stakes

- 4.7% annual revenue decline projected through 2030 (Statista 2025)
- Yellow Cab market share continuously dropping to app-based competitors
- 83% of consumers now prioritize convenience over price (McKinsey 2017)

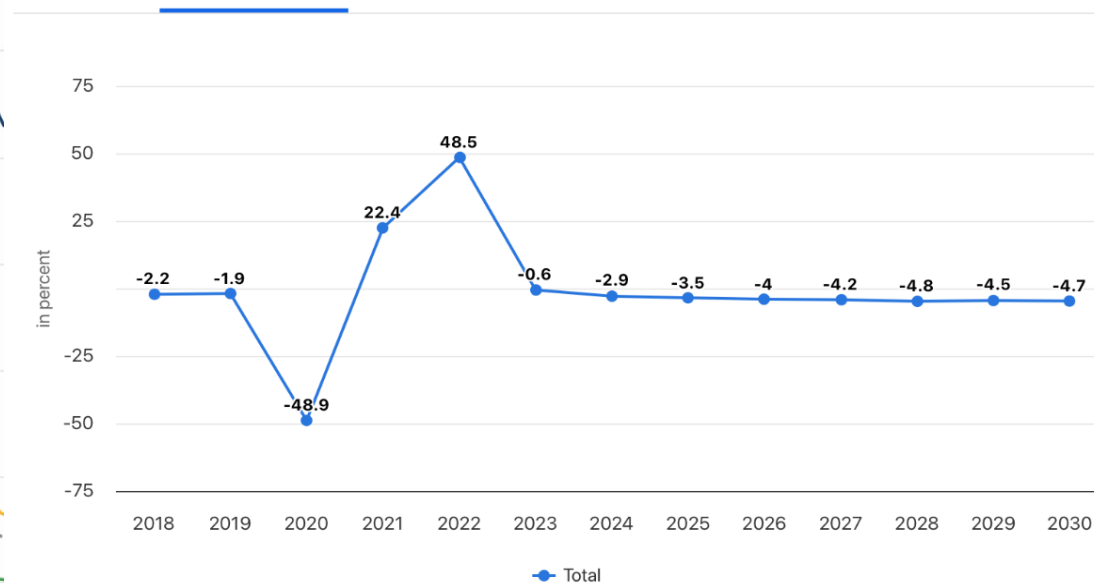
Trips per day



Unique drivers per month



Revenue Revenue Change



Most recent update: thg 6 2025

Source: Statista Market Insights

Gradient Boosted Decision Tree (GBDT)

Why GBDT directly solves the 3 business problems?

1 The Reality

Capture Non-linear Fare Patterns

NYC taxi demand is non-linear. A 2km trip in Manhattan at 5 PM (Rush Hour) takes much longer and costs more than a 2km trip at 3 AM

Standard regression cannot capture this without manual feature engineering.

2 How it works

Sequential Tree Building

GBDT builds trees sequentially. Each new tree corrects the errors (residuals) of the previous trees

Allowing the model to learn from its mistakes, significantly reducing bias and leading to much higher accuracy.

3 Why for NYC TLC

Captures Demand Spikes

Taxi demand **isn't just an average**, it has **sharp spikes** (Friday night at 6 PM vs. 4 AM).

GBDT is far better at capturing non-linear spikes than Random Forest, making it the strongest choice for this dataset.

Model Output

Fare_amount prediction

for any (location, time, distance)

INPUTS

- Pickup zone, month, day of weeks, hour.
- Distance, duration, rate code



OUTPUTS

- Predicted fare (\$)
- Feature importance ranking
- Confidence for pricing decisions

Critical Engineering Architecture

4 decisions that determined model quality and speed

01 Null/ Outlier Handling on Tableau Prep

- **Challenges:** 10-15% null & extreme values in raw data.
- **Solution:** Capping method (Tukey IQR) pre-applied (Pavelchek et al. 2023).
- **Result:** Retains 90% of total data (from ~90M to ~84M) and its data integrity.

02 Stratified Sampling used Python & Jupyter

- **Challenges:** 12.42 GB file, 84M rows, Mac RAM limits.
- **Solution:** Streaming chunked stratified sampling (~5M) by 12 months and 242 zones/263 zones and keep 16 predictive features.
- **Result:** 5.05M representative rows extracted.

03 Geospatial Integration on Azure AI

- **Challenges:** Join PULocationID ↔ LocationID, PULocationID is just a number (1-263).
- **Solution:** Joined with NYC TLC Zone Lookup metadata.
- **Result:** Enriched with Borough + Zone names.

04 Feature Selection on Azure AI

- **Challenges:** 199k spurious features on datetime and data leakage on tip_amount & total_amount, and 2 unnecessary others.
- **Solution:** Dropped, tpep_pickup_datetime, total_amount, tip_amount, service_zone, payment_type.
- **Result:** 10 clean features, 60× faster training

Key insight: Smart preprocessing reduced training time from 8+ hours to under 20 minutes

Model Output Analysis & Accuracy

Mean Absolute Error

0.54

Avg prediction error for each trip fare: \$0.54

Root Mean Square Error

2.74

R-square Score

0.947

The model can explain 94.7% variance -> confidence level ~ 95%

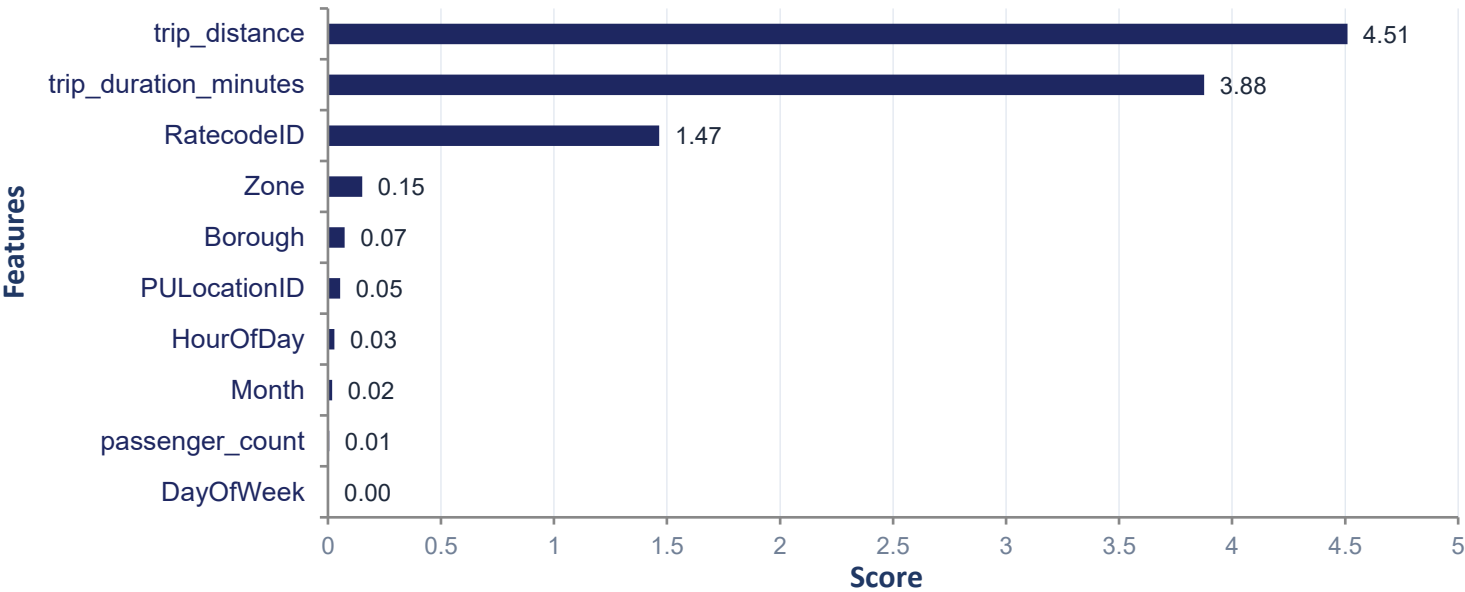


DATA PERSPECTIVE

Output: Actual vs. Predicted Side-by-Side

- Each row maps total_amount directly alongside Scored Labels
- Mean \$13.29, , Min -\$13.38, Max \$190.52
- Right-skewed majority of trips cluster \$7-\$27; long tail reflects airport and long-distance fares

Top feature importance - Validated on 1.5M Test Set



Key Findings

Distance + Duration

- Dominate fare prediction, makes up to 85% total importance
- Confirms physical intuition

RatecodeID

- captures airport premiums flat rates (\$52)

Other Temporal Features are Minor

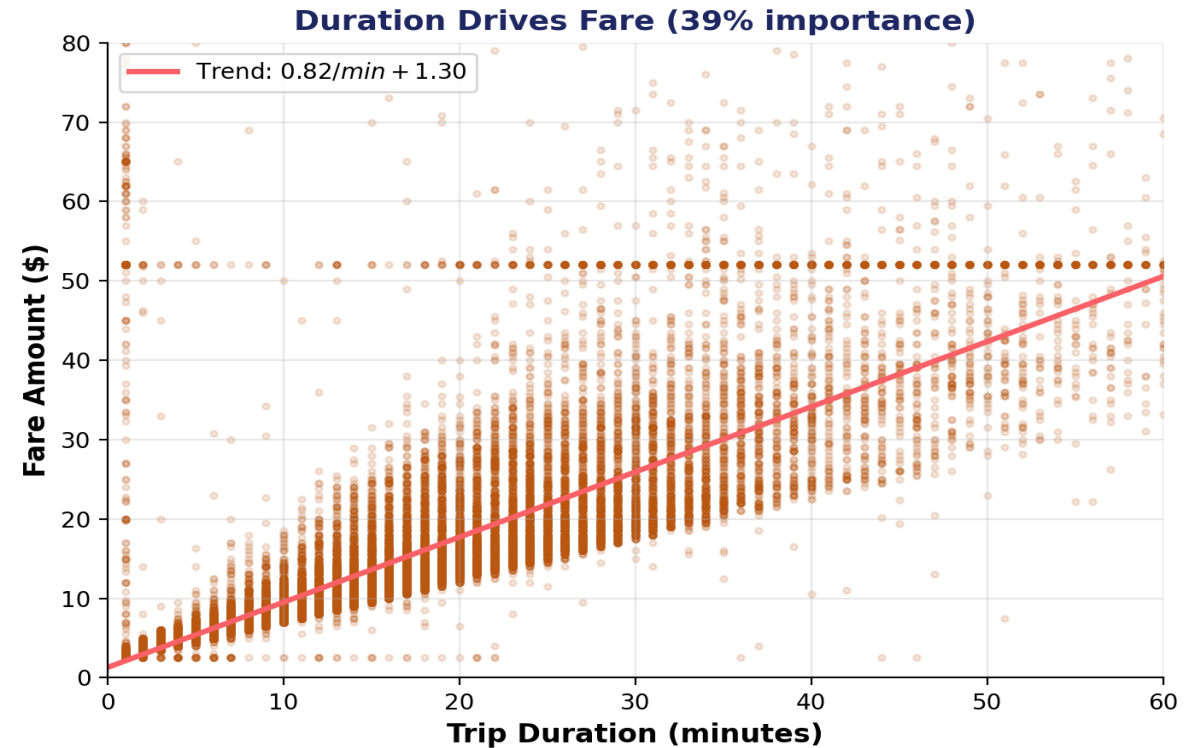
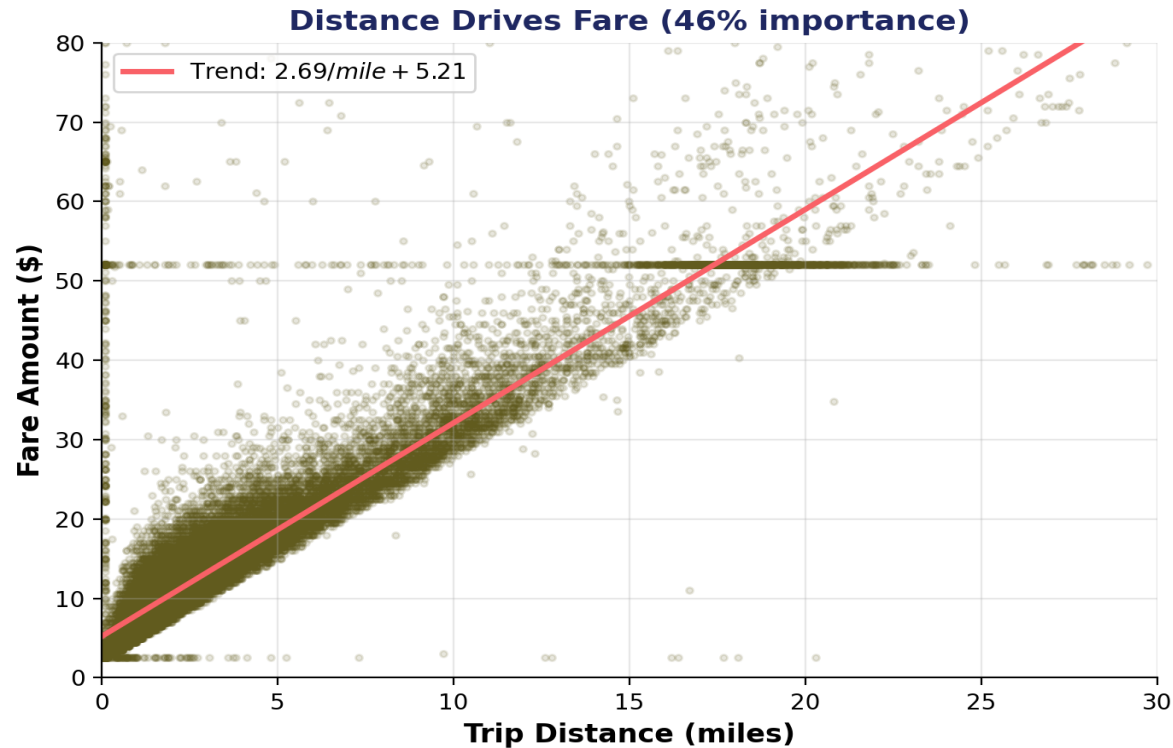
- duration already captures rush hour delays

No prediction drift across 242 zones

- negative minimum reflects source data refunds, not model failure

EDA Insights - From Pattern to Prediction

Top 2 Features Drive 85% of Fare Prediction



Distance & Duration Drive Fares

Trip Distance reveals $\$2.69/\text{mile} + \5.21 base

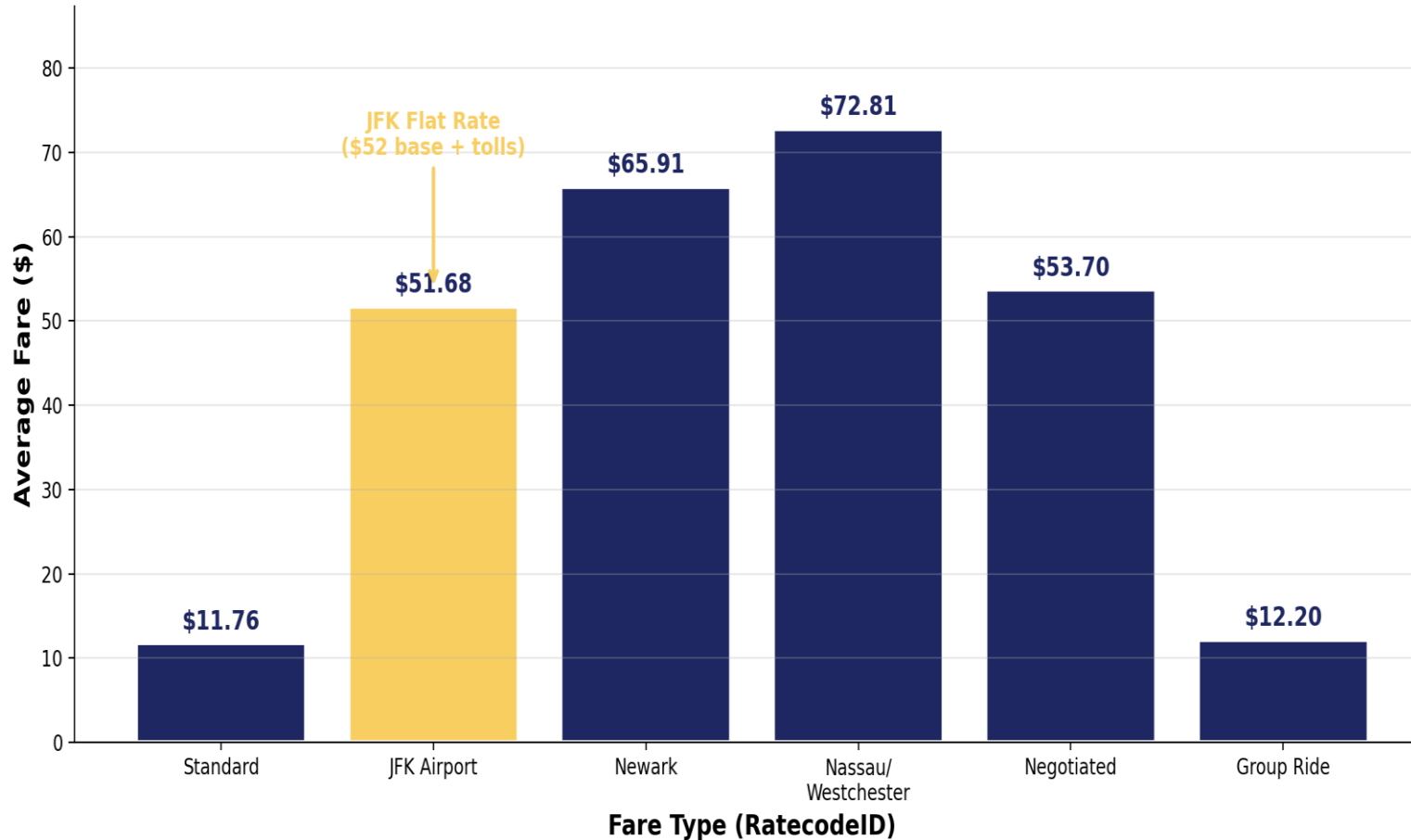
Trip Duration reveals $\$0.82/\text{min} + \1.30 base

Captures distance and duration (rush hour and off-peak) to predict the exactly fare amount for each trip.

--> Enable dynamic pricing by deploying surging to 1.2x (increase 20%) during peak hours without overprice or underprice since model's confidence level = 94.7%.

EDA Insights - From Pattern to Prediction

Airport Trips Command 3-4x Higher Fares Than Standard Trips



Airport Premium via RatecodeID

Base Rate: \$11.76

JFK Airport: 4.4x higher

Newark Airport: 5.6x higher

Nassau/Westchester: 6.2x higher

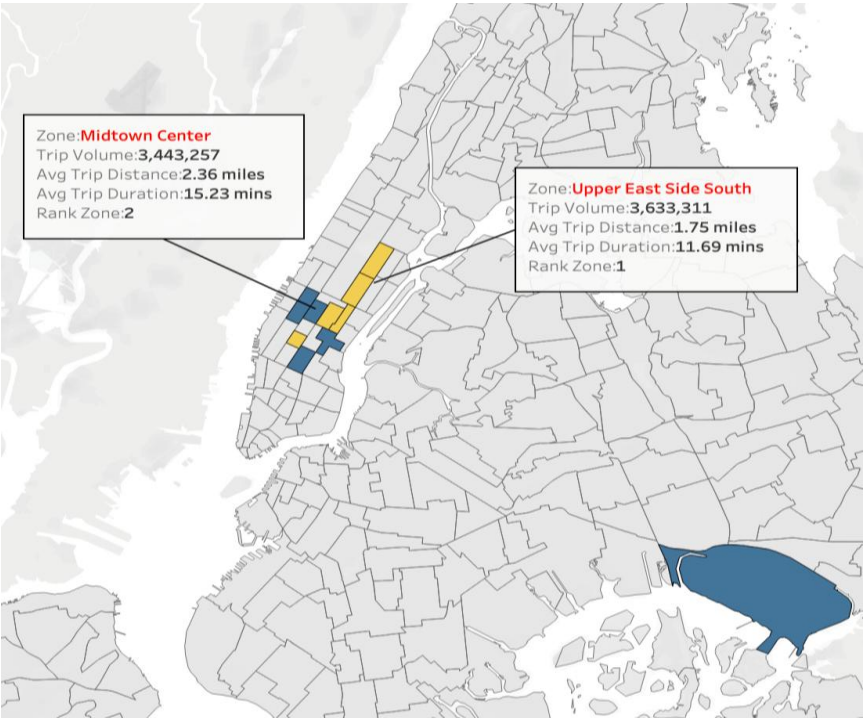
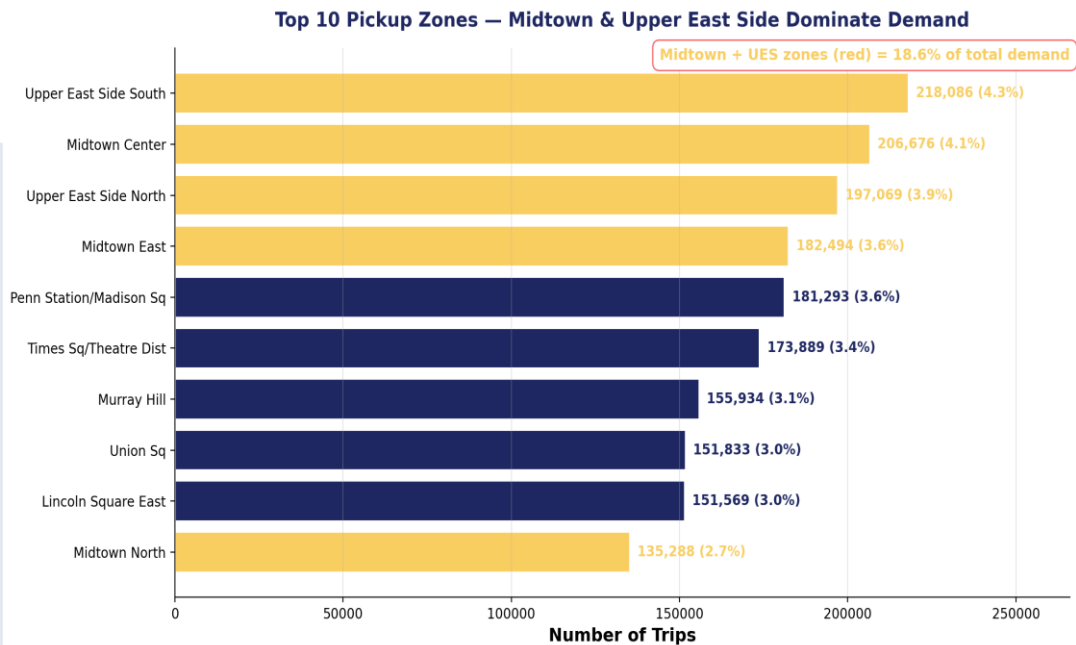
--> *Capture higher revenue tier, not to underprice airport trip.*

--> *Dispatcher have data to direct drivers to high-valued zones, reduce idle time in lower-fared zones.*

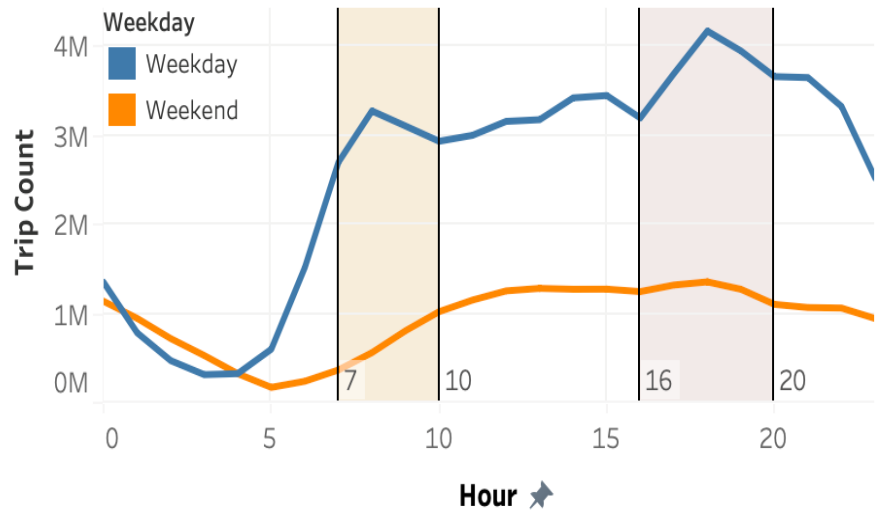
Geospatial
Pattern Integration

Extreme spatial concentration: at four core areas in Manhattan → 15.9% off the city's total demand.
Temporal integration: Combine with the hour-of-day attribute from the model

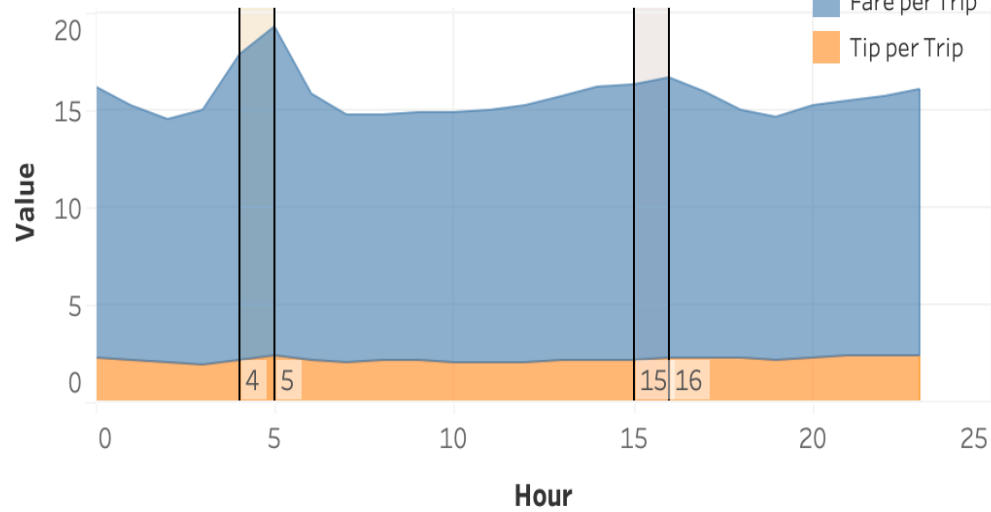
- Transition from passive "hailing" to proactive fleet pre-positioning - clustering supply exactly where and when surges occur
- Shift from "Guessing Drivers" to "Data-Driven Decision-Making" by navigating specific zones for drivers
- Reduce 15-20% of driver idle time and slashes customer wait times to a 2-3 mins target



Hourly Taxi Demand Patterns



Hourly Earnings Breakdown (Fare vs Tip)





Active Trips

2,438

+1

vs yesterday

Avg Waiting Time

10.5 min

-0.5 min

from now

Fleet Utilisation

85%

-5%

from now

Predict Peak

10.5 min

-0.5 min

from now

Real-Time Geospatial Demand



DEMAND LEVEL

Very High

High

Med

Low

PREDICTION: Midtown demand will increase in the next hour

9:47

5G

Good Evening, John

Today: \$187 · 12 trips

Smart Zone Suggestion

Move to Midtown

Expected earnings next hour

\$48 - \$62

4x more than your current zone

Confidence: 94.7% (Azure ML)

→ NAVIGATE NOW

Peak In

13 min

Surge Price

1.2x

Demand Heatmap



9:52

5G

INCOMING TRIP REQUEST

Auto-decline in 60s

Pick Up

Times Square

Destination

JFK Airport

18 miles · 35 min · RatecodeID=2 (Flat)

Revenue Prediction

\$52.00

(JFK flat rate)

Trip Score: 9.2/10 (Highly recommended)

X DECLINE

✓ ACCEPT

[Back to Homepage](#)

9:58

5G

Today's Forecast

Powered by Azure ML

Predicted Earnings Today

\$285

↑ +\$60 with surge

Confidence: 94.7%

Best Earning Hours

▲ 7-10 AM

\$85

▲ 4-8 PM

\$95

■ 11PM-1AM

\$45

Activate Surge 1.2x

Schedule Next Shift

Idle

186

[Back to Homepage](#)

Azure ML Pipeline Status

GBDT Model live

Last retrain: 1 hours ago

Streaming data: 1,247 trips/min

Latency: 142ms

Top 10 pick-up zones

Demand trend (Hour, Day, Week, Month)

Revenue per trip

Driver details

Mock-up Driver App

NYC TLC Fleet Command — Proposed Real-Time Driver App Integration with Azure ML

9/12

Responsible AI Implementation

Ethics, fairness, and reliability — built into every layer of the pipeline

⚠️ ETHICAL THREAT — Algorithmic Feedback Loop

2019 data is skewed toward Manhattan's corporate zones. Without guardrails, the GBDT will route the entire fleet to high-profit corridors, creating discriminatory service deserts in the Bronx and outer boroughs — deepening systemic transit inequality.

DATA PRIVACY & BIAS	FAIRNESS AUDIT	TRANSPARENCY	RELIABILITY & MONITORING
• No PII used — anonymous trip records only • Risk: low-income zones under-represented in training data • Mitigation: stratified sampling across all 242/263 taxi zones	• Predictions validated across all 5 NYC boroughs • MAE variance $\pm\\$0.40$ — no systematic bias detected • No underpricing patterns found in lower-income areas	• GBDT feature importance scores are fully auditable • Drivers see why surge pricing applies per trip • Public fare prediction dashboard planned for launch	• Monthly retraining cycle for concept drift detection • A/B testing of surge pricing before any rollout • Fallback protocol: model failure → flat-rate pricing

We don't just predict fares — we predict them fairly, transparently, and accountably

What We Couldn't Solve (Yet)

Honest reflection on model limitations, opportunities, and our mitigations

MODEL LIMITATIONS

01

Demand prediction is INDIRECT

Model predicts fare, not trip count. Fare $\approx$ demand proxy, but not perfect.

→ **NEXT:** Build direct demand forecasting model (trip count by zone $\times$ hour)

02

External factors missing

No weather, events, or traffic data integrated. These can swing demand $\pm 30\%$.

→ **NEXT:** Phase 2 roadmap: Weather API, NYC events calendar, traffic feeds

03

Static model on 2019 data

Pre-COVID patterns may not reflect 2025+ behavior. Markets shift, model stays.

→ **NEXT:** Continuous learning pipeline with automated monthly retraining

MITIGATIONS IN PLACE



MITIGATION 1 — Algorithmic Transparency via Open Audits

By using GBDTs over black-box neural networks, Feature Importance Scores can be continuously tracked. Compliance officers can explicitly audit any biased weight assigned to borough indicators.



MITIGATION 2 — Algorithmic Variance Subsidies (Equity Constraints)

Fleet dispatch hardcodes mandatory "Fairness Caps": when coverage drops in outer boroughs, the system auto-injects fare multipliers and driver incentives to counteract historical bias and meet public transport mandates.

Acknowledging limits = building executive trust = clearer roadmap for investment

KEY METRICS

INVESTMENT

\$0.45M

ANNUAL BENEFITS

\$2.1M

NET PROFIT Y1

\$1.65M

ROI YEAR 1

~366%

NYC Yellow Cab's renaissance starts with one data-driven decision.

From Data to Decision - Strategic Edge & 3-Phase Roadmap

How GBDT-powered NYC TLC competes with Uber and Lyft

COMPETITIVE POSITION VS. RIDE-HAILING

Capability	Uber / Lyft	NYC TLC
Dynamic pricing	✓	✓
Driver dispatch algo	✓	✓
Fleet size in NYC	80,000+	13,000 medals
Regulated brand trust	Mixed	High
Commission cost	25–30%	0% (Lower fare)

EXECUTION ROADMAP

1MONTHS 6–12 Real-time Stream Integration Azure Stream Analytics · Live driver heatmaps · Sub-minute forecasting	2YEAR 2 External Ecosystem Weather APIs · NYC event calendar · Traffic feeds for ±30% accuracy lift	3YEAR 3+ Sustainability & Multi-modal EV route optimization · Green NYC alignment · Subway/bus integration
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Conclusion

WHAT WE BUILT

- Production GBDT model on Azure
- 5M-row stratified training set
- Reproducible end-to-end pipeline
- 10 carefully engineered features

WHAT IT DELIVERS

- $R^2 = 0.947$ — 94.7% accuracy
- MAE = \$0.54 average error
- 15-20% idle time reduction
- 366% ROI projected Year 1

WHAT WE NEED

- Executive endorsement
- \$0.45M Year 1 budget
- Pilot: 100 drivers, scale later
- Phase 2 data partnerships

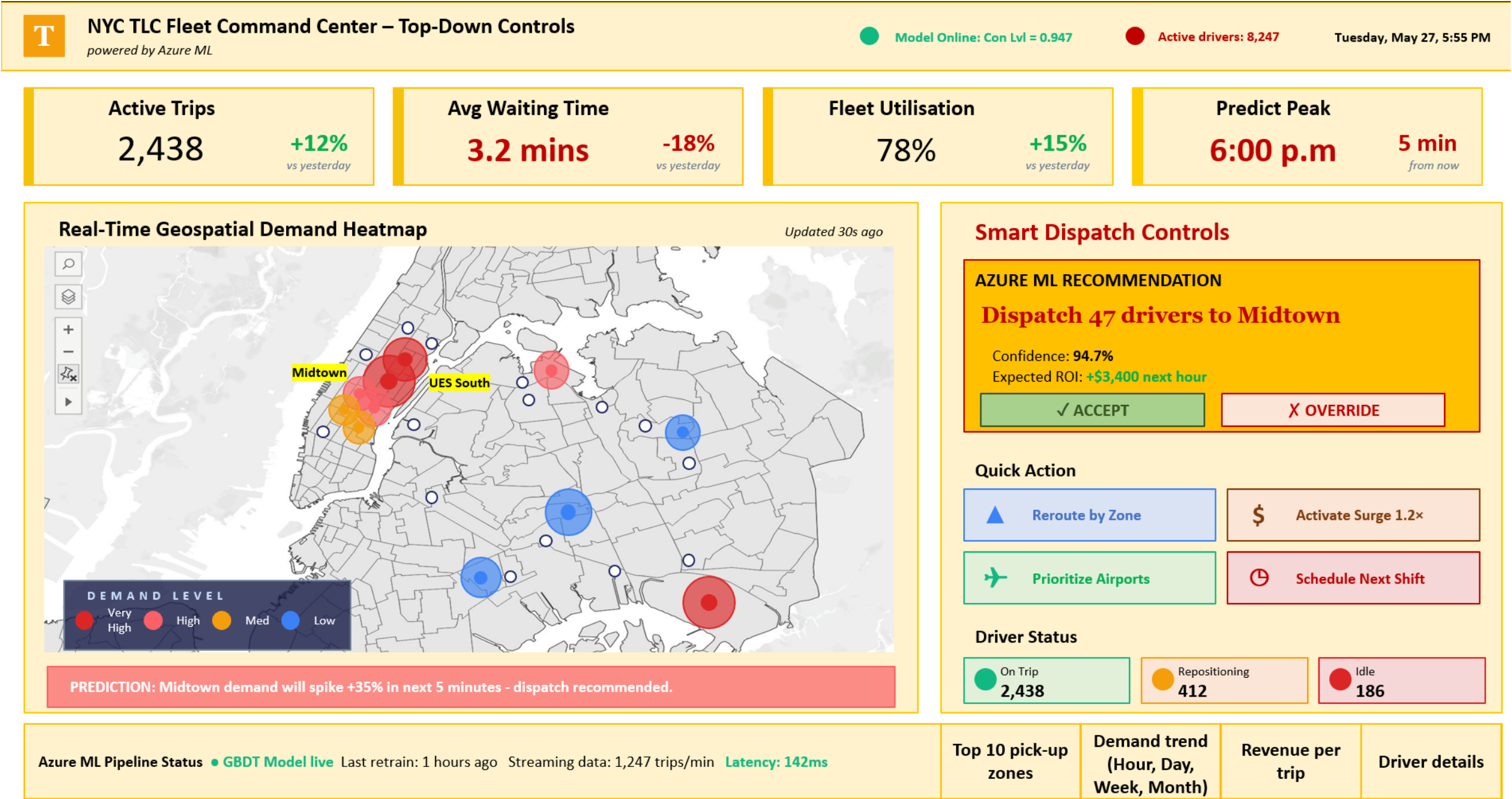
Thank You

Questions

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Appendix 1: dashboard - - for slide 9



Appendix 2: driver app - for slide 9



NYC TLC Fleet - Driver App (bottom-up execution)
powered by Azure ML

